

SHORT TANDEM REPEAT (STR) MOLECULAR HEMATOPOIETIC CHIMERISM

Date: September 11, 2026

Patient Name:	Aarav Mukherjee	MRN / Patient ID:	FMRI - BMT - 2026 - 118
Age / Gender:	16 Yrs / Male	Blood Group & Rh:	O+ (Post-Transplant Recipient)
Clinical Diagnosis:	Acute Lymphoblastic Leukemia (T-ALL Post Day +60)	Referring Consultant:	Dr. Rahul Bhargava, MD, DM

Fraction Analyzed	Donor Specific Alleles	Recipient Specific Alleles	Engraftment Status
Total Nucleated Cells (Whole Blood)	99.8% Donor	0.2% Recipient	Full Donor Chimerism
T-Lymphocyte Fraction (CD3+ Cells)	99.4% Donor	0.6% Recipient	Complete T-Cell Engraftment
Myeloid Granulocyte (CD33+ Cells)	100.0% Donor	0.0% Recipient	100% Donor Lineage
B-Lymphoid Fraction (CD19+ Cells)	99.6% Donor	0.4% Recipient	Full Donor Chimerism

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- STR analysis of 16 informative polymorphic loci confirmed complete donor chimerism (> 99.0%) across all lineages.
- Zero evidence of recipient hematopoietic relapse or graft failure at Day +60 post allogeneic HSCT.
- Recommendation: Maintain current cyclosporine immunosuppression taper and continue bi-weekly molecular surveillance.

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